

HR Analytics: Job Change Prediction

- **Binary classification problem**
- **Predict whether a candidate is likely to change jobs**
- **Dataset: HR Analytics Job Change of Data Scientists**
- **Target variable: target**
- **70% train / 15% validation / 15% test**

Preprocessing and Machine Learning Models

Preprocessing

- **Missing value imputation**
- **Encoding categorical variables**
- **Feature scaling**
- **Feature engineering**
- **Reused preprocessing pipeline from the midterm project**

Models

- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Gradient Boosting**
- **KNN**
- **SVC**
- **Voting Ensemble**
- **Bayesian Ensemble**

	Model	Dataset	Accuracy	Precision	Recall	F1 Score	ROC-AUC
7	Gradient Boosting	Test	0.7756	0.5661	0.4246	0.4852	0.7891
13	Voting Ensemble	Test	0.7791	0.5595	0.5321	0.5455	0.7873
5	Random Forest	Test	0.7735	0.5455	0.5447	0.5451	0.7837
15	Bayesian Ensemble	Test	0.7582	0.5175	0.4330	0.4715	0.7412
1	Logistic Regression	Test	0.7074	0.4405	0.6466	0.5241	0.7334
9	K-Nearest Neighbors	Test	0.7491	0.4958	0.4106	0.4492	0.7132
11	Support Vector Classifier	Test	0.7220	0.4446	0.4651	0.4546	0.6852
3	Decision Tree	Test	0.7227	0.4444	0.4525	0.4484	0.6320
6	Gradient Boosting	Validation	0.7888	0.6054	0.4407	0.5101	0.8061
12	Voting Ensemble	Validation	0.7898	0.5879	0.5272	0.5559	0.7992
4	Random Forest	Validation	0.7878	0.5790	0.5467	0.5624	0.7950
14	Bayesian Ensemble	Validation	0.7672	0.5424	0.4282	0.4786	0.7455
0	Logistic Regression	Validation	0.7122	0.4462	0.6360	0.5244	0.7321
8	K-Nearest Neighbors	Validation	0.7512	0.5018	0.3849	0.4357	0.7210
10	Support Vector Classifier	Validation	0.7370	0.4734	0.4840	0.4786	0.6923
2	Decision Tree	Validation	0.7161	0.4298	0.4226	0.4262	0.6181

Results and Best Model

Best Model
Gradient Boosting Classifier

Validation ROC-AUC: 0.8061
Test ROC-AUC: 0.7891

The Voting Ensemble achieved competitive performance, but Gradient Boosting produced the best overall ROC-AUC results

Conclusion

- **Successfully implemented a complete machine learning classification workflow**
- **Compared multiple classification and ensemble models**
- **Gradient Boosting achieved the best performance**
- **Demonstrated the importance of preprocessing and model evaluation**